

Apoorv Vuppulury

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Summary

Final-year ECE student with a strong focus on computer architecture and hardware–software co-design, backed by a solid foundation in embedded systems, VLSI design, and system-level modeling. Experienced in embedded control and timing/performance analysis across hardware–software boundaries, with a proven record of executing complex, long-term technical & interdisciplinary projects.

Education

B.E. in Electronics and Communication Engineering	Oct, 2022 – Jul, 2026
BITS Pilani - Hyderabad Campus	CGPA: 8.01/10.00
• Coursework: Computer Architecture, Digital VLSI Design, Machine Learning, Embedded System Design, Digital Design • NPTEL: Computer Architecture and Organization, Hardware Modeling using Verilog, 5G - Wireless Standard Design	

Experience

Firmware Engineer - II	Jul, 2026 - Present
<i>Dhruva Space</i>	Hyderabad, TG
• Part of Satellite IoT & Data Relay Systems to support the design and deployment of SmallSat-based IoT data relay solutions. Working on end-to-end IoT systems across space and ground segments and the ability to translate customer use cases into practical satellite-IoT architectures.	
Embedded AI Intern	Jan, 2026 - Present
<i>Inferigence Quotient</i>	Bengaluru, KA
• Designed and implemented a cascaded closed-loop 3-axis gimbal control system using Teensy 4.x, featuring interrupt-driven encoder decoding and real-time PID gain tuning for deterministic, low-latency position control. • Developed a MATLAB/Simulink-based dynamic model of the control loop and validated hardware performance through flight telemetry analysis (QGC), achieving <5% deviation between simulation and experimental results.	

Projects	
Asynchronous FIFO Memory System GitHub Repository	Apr, 2025 - Oct, 2025
• Designed a clock-domain-crossing (CDC)-based asynchronous FIFO with Gray-coded pointers and dual flip-flop synchronizers to safely transfer data between asynchronous clock domains and mitigate metastability. • Developed an FSM-based request–acknowledge handshake to enable reliable single-word transfers across the asynchronous buffer.	
5 - Stage pipeline RISC Processor GitHub Repository	Jan, 2025 - Apr, 2025
• Designed a 5-stage pipelined RISC CPU, supporting custom (R,I,J-type) instruction opcodes with complete datapath and control logic. Verified pipeline correctness through waveform-based simulation in Xilinx Vivado. • Implemented hazard detection and data forwarding units to resolve pipeline data hazards and minimize stall cycles.	
ML and QoS-Aware Routing for LEO Satellite Networks 	Apr, 2025 - Nov, 2025
• Designed a QoS-aware routing framework for dynamic LEO inter-satellite networks using Graph Neural Networks (GNNs) for topology embedding and Deep Q-Networks (DQN) for adaptive path selection under time-varying link conditions. • Achieved up to 35% latency reduction and improved QoS satisfaction by 8% compared to Dijkstra and CGR baselines.	
Embedded Edge-Framework for Autonomous Collision Avoidance 	Aug, 2025 - Dec, 2025
• Developed an embedded vision platform on Raspberry Pi and NVIDIA Jetson Nano with cameras for onboard autonomous obstacle detection and collision avoidance. Implemented an OpenCV-based image processing pipeline to detect obstacles from live video and trigger evasive action.	
Resource Allocation in STAR-RIS Assisted Wireless 6G Networks 	Oct, 2024 - Dec, 2025
• Investigated resource allocation strategies, deriving mathematical bounds and critical trade-offs on element allocation factors and phase error constraints to optimize spectral efficiency. Developed a three-stage framework for intelligent coverage hole detection and remediation in PPP networks that achieved 100% coverage, while improving baseline coverage by 12.28%.	

Technical Skills

• Tools: Xilinx Vivado, MATLAB (Simulink, Simscape), Arduino IDE, STM32Cube IDE, Keil MDK-ARM • Languages: Verilog HDL, ARM Assembly, MATLAB, C, C++, Python (NumPy, SciPy, TensorFlow, Pandas, Matplotlib) • Communication Protocols: UART, SPI, I2C, MavLink, Ethernet, WIFI (IEEE 802.11), HTTP, TCP, UDP • Embedded Platforms: Arduino, ESPs, Teensy 4.x, Raspberry Pi, STM32, NVIDIA Jetson Nano, FOC Shields	
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Positions of Responsibility

Founding Member, Training and Counseling Lead Student Welfare and Mentorship Committee	May, 2024 – Aug, 2025
• Designed the first comprehensive mentorship program of BPHC, and led a team of 150 student mentors, positively impacting over 500 first-year students and addressing key transitional challenges from Day 0. • Served as a member of the Student–Faculty Council and received a Certificate of Appreciation for contributions to the Dept. of EEE.	